1 Background

Standard Differential Evolution (Greedy Selection)

In standard DE, a trial vector $\mathbf{u}_i^G$ generated via mutation and crossover replaces its target vector $\mathbf{x}_i^G$ in the next generation $G + 1$ **only if** it is equal or superior in fitness:

$$\mathbf{x}_i^{G+1} = \begin{cases} \mathbf{u}_i^G, & \text{if } f(\mathbf{u}_i^G) \leq f(\mathbf{x}_i^G) \\ \mathbf{x}_i^G, & \text{otherwise} \end{cases} \tag{1}$$

While effective, this strictly greedy replacement strategy can cause the population to rapidly lose diversity, leading to **premature convergence** or **stagnation** on multimodal or deceptive fitness landscapes.

DE Variants

Adaptive Differential Evolution with Optional External Archive (JADE) JADE enhances classic Differential Evolution by introducing the DE/current-to-$p$best/1 mutation strategy alongside an optional external archive A. Instead of using fixed control parameters, JADE dynamically updates scale factors F_i and crossover rates CR_i at each generation. The scale factors are sampled from a Cauchy distribution $\mathcal{C}(\mu_F, 0.1)$, while the crossover rates are drawn from a Normal distribution $\mathcal{N}(\mu_{CR}, 0.1)$. The location parameters μ_F and μ_{CR} are continuously adapted based on historical success rates of parameters that successfully generated superior offspring. Furthermore, retaining recently discarded parent vectors in the external archive maintains population diversity and prevents premature convergence on complex multimodal landscapes. —–

Table 1: Adaptive Differential Evolution with Optional External Archive (JADE)

Variant	**Primary Reference / Citation**	**Publication Venue**	**DOI / Source**
JADE	Zhang, J., & Sanderson, A. C. (2009). *JADE: Adaptive Differential Evolution with Optional External Archive.*	*IEEE Transactions on Evolutionary Computation*, 13(5), 945–958.	10.1109/TEVC.2009.2014613

Success-History-Based Adaptive Differential Evolution (SHADE) SHADE extends the foundational concepts of JADE by removing reliance on static probability distributions for parameter generation. Instead, SHADE maintains a historical memory set $H = \{(M_{CR,k}, M_{F,k})\}_{k=1}^{H}$ containing H pairs of successful crossover rates and scale factors. During each generation, parameter values are generated by randomly selecting an entry index k from memory, with CR drawn from a Normal distribution $\mathcal{N}(M_{CR,k}, 0.1)$ and F from a Cauchy distribution $\mathcal{C}(M_{F,k}, 0.1)$. The historical memory is dynamically updated at the end of each generation using the weighted Lehmer mean for scale factors, and the weighted arithmetic mean for crossover rates based on fitness improvements. This memory-driven adaptation mechanism provides superior robustness across complex, non-linear optimisation problems.

Table 2: Success-History-Based Adaptive Differential Evolution (SHADE)

Variant	**Primary Reference / Citation**	**Publication Venue**	**DOI / Source**
SHADE	Tanabe, R., & Fukunaga, A. (2013). *Success-history based parameter adaptation for Differential Evolution.*	*IEEE Congress on Evolutionary Computation (CEC)*, 71–78.	10.1109/CEC.2013.6557555

Linear Population Size Reduction SHADE (L-SHADE) L-SHADE improves upon the performance of SHADE by incorporating a deterministic Linear Population Size Reduction (LPSR) scheme. Recognising that high population sizes are beneficial during initial exploration but computationally wasteful during fine-grained exploitation, L-SHADE dynamically decreases the population size N linearly from $N_{\max}$ to $N_{\min}$ over the course of the maximum allowed number of function evaluations (MAX_NFE). When the population size shrinks, worst-performing individuals are removed, while the remaining parameter history adaptation follows the SHADE framework. This dynamic balance between early global exploration and late local refinement enabled L-SHADE to achieve winning performances across multiple IEEE Congress on Evolutionary Computation (CEC) benchmark competitions.

Table 3: Linear Population Size Reduction SHADE (L-SHADE)

Variant	Primary Reference / Citation	Publication Venue	DOI / Source
L-SHADE	Tanabe, R., & Fukunaga, A. (2014). *Improving the search performance of SHADE using linear population size reduction.*	*IEEE Congress on Evolutionary Computation (CEC)*, 1658–1665.	10.1109/CEC.2014.6900380

Advanced L-SHADE Extensions (iL-SHADE, L-SHADE-EpSin, and L-SHADE-RSP) Following the success of L-SHADE, several high-performance variants were developed to address specific optimisation challenges:

- **L-SHADE-EpSin** incorporates an ensemble of sinusoidal formulas for adapting the scale factor F, allowing local search and global exploration phases to alternate cyclically according to sinusoidal trajectories over iterations.
- **L-SHADE-RSP** integrates Rank-Space Perturbation, which modifies vector selection probabilities in mutation operators using individual rank positions in the current population rather than purely uniform random selection.

Collectively, these extensions demonstrate state-of-the-art performance when tackling high-dimensional, non-separable, and rotated benchmark landscapes.

Table 4: Literature References for Adaptive and Self-Adaptive DE Variants

Variant	Primary Reference / Citation	Publication Venue	DOI / Source
L-SHADEExtensions	Brest, J., et al. (2017). *Single objective real-parameter optimisation (iL-SHADE).* Stanovov, V., et al. (2018). *L-SHADE with rank-based selective perturbation (L-SHADE-RSP).*	*IEEE Congress on Evolutionary Computation (CEC) / Information Sciences*	10.1109/CEC.2017.7969440 10.1016/j.ins.2018.03.061

2 Overview

Late Acceptance Differential Evolution is a hybrid metaheuristic variant that incorporates the **Late Acceptance (LA)** move-acceptance strategy (originally introduced by Burke and Bykov in 2008 for Late Acceptance Hill Climbing) into the **Differential Evolution (DE)** algorithm framework.

It relaxes DE's strict, greedy selection mechanism to prevent premature convergence and help the search escape local optima in complex search landscapes.

3 Background & Motivation

4 How Late Acceptance Works

The **Late Acceptance** strategy modifies the selection decision by comparing a new trial solution's quality against a **historical benchmark** rather than just the immediate parent or current solution.

1. **Fitness History Array**: A circular list or history buffer H of fixed length L maintains fitness values from past generations or iterations.
2. **Acceptance Rule**: A trial candidate $\mathbf{u}_i^G$ is accepted if its fitness $f(\mathbf{u}_i^G)$ is better than (or equal to) either:
   - The fitness of the current target vector $f(\mathbf{x}_i^G)$, **OR**
   - The fitness stored L steps ago in the history buffer: $H[G \bmod L]$.

4.1 Selection Formula

$$\mathbf{x}_i^{G+1} = \begin{cases} \mathbf{u}_i^G, & \text{if } f(\mathbf{u}_i^G) \leq f(\mathbf{x}_i^G) \quad \text{or} \quad f(\mathbf{u}_i^G) \leq H[G \bmod L] \\ \mathbf{x}_i^G, & \text{otherwise} \end{cases} \tag{2}$$

After evaluating the decision, the current fitness (or best fitness) is written back into $H[G \bmod L]$.

5 Key Benefits

- **Escaping Local Optima**: Because $H[G \bmod L]$ reflects an older, often worse fitness state from L generations prior, trial vectors that are temporarily worse than the current generation can still be accepted. This allows the search trajectory to jump over local fitness barriers.
- **Minimal Parameter Tuning**: Unlike Simulated Annealing (which requires tuning cooling schedules, initial temperatures, and decay rates), Late Acceptance relies on a single intuitive parameter: the history length L.
- **Population Diversity Maintenance**: Accepting non-improving steps periodically injects structural diversity into the DE vector population, delaying stagnation.